

# **CSCE 775: Reinforcement Learning**

Practice Quiz

Chapters 1, 3, 4

J.C. Vaught

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## **Instructions**

This quiz covers material from Chapters 1, 3, and 4. Show all work and justify your answers.

**Problem 1** (Computing Returns with Discount Factors). *Given a discount factor  $\gamma = 0.9$  and a reward sequence:  $R_1 = 2$ ,  $R_2 = -1$ ,  $R_3 = 3$ ,  $R_4 = 5$ ,  $R_5 = 0$  with terminal time  $T = 5$ :*

- (a) Compute  $G_0, G_1, G_2, G_3, G_4, G_5$  by working backwards from  $G_5 = 0$ .*
- (b) If  $\gamma = 0.5$ , what are  $G_0$  through  $G_5$ ?*

**Problem 2** (Discounted Return for Infinite Sequence). *Given  $\gamma = 0.8$  and a constant reward of  $+3$  at every time step (continuing task):*

- (a) What is the return  $G_t$  for any time  $t$ ?*
- (b) If the first reward is  $+10$  and all subsequent rewards are  $+3$ , what is  $G_0$ ?*

**Problem 3** (Bellman Equation Verification on Small MDP). *Consider a 3-state MDP with states  $\{A, B, C\}$  and actions  $\{a_1, a_2\}$ :*

**Transition dynamics:**

- *From A:*
  - $a_1$ : go to B with prob 0.7 (reward +2) or stay at A with prob 0.3 (reward +1)
  - $a_2$ : go to C with prob 1.0 (reward +5)
- *From B:*
  - $a_1$ : go to A with prob 0.4 (reward +3) or C with prob 0.6 (reward -1)
  - $a_2$ : stay at B with prob 1.0 (reward +2)
- *From C:*
  - $a_1$ : go to A with prob 1.0 (reward +4)
  - $a_2$ : go to B with prob 0.5 (reward +1) or C with prob 0.5 (reward -2)

**Policy:**  $\pi(a_1|A) = 0.7$ ,  $\pi(a_2|A) = 0.3$ ;  $\pi(a_1|B) = 0.5$ ,  $\pi(a_2|B) = 0.5$ ;  $\pi(a_1|C) = 0.6$ ,  $\pi(a_2|C) = 0.4$

With  $\gamma = 0.9$ :

- (a) Draw the transition diagram.
- (b) Set up the Bellman equations for  $v_\pi$ .
- (c) Solve the system of linear equations for  $v_\pi(A)$ ,  $v_\pi(B)$ ,  $v_\pi(C)$ .

**Problem 4** (Policy Evaluation on a Gridworld). *Consider a  $3 \times 3$  gridworld with the terminal state at the top-right corner. Reward =  $-1$  on all transitions.  $\gamma = 1.0$  (undiscounted episodic task).*

- (a) Draw the gridworld with TikZ.*
- (b) Using a uniform random policy (all 4 actions equally likely), perform 3 iterations of iterative policy evaluation by hand.*
- (c) Start with  $V(s) = 0$  for all states.*
- (d) Actions that would take the agent off the grid keep the agent in place.*
- (e) Show the value table after each sweep.*

**Problem 5** (Policy Improvement). *Using the converged values from Problem 4, determine the greedy policy for each state. For each non-terminal state, show which action(s) maximize  $q_\pi(s, a)$ . Draw the resulting greedy policy with arrows.*

**Problem 6** (Value Iteration Steps). *Using the same  $3 \times 3$  gridworld from Problem 4, apply value iteration:*

$$V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)[r + \gamma V(s')]$$

*Perform 3 sweeps of value iteration starting from  $V(s) = 0$  for all states. Show the value table after each sweep and determine the optimal policy from the final values.*

**Problem 7** (Identifying Optimal Policies). *Consider a 2-state MDP with states  $\{s_1, s_2\}$  and  $\gamma = 0.9$ :*

***Transition dynamics:***

- *From  $s_1$ : action ‘stay’  $\rightarrow s_1$  with reward +1; action ‘go’  $\rightarrow s_2$  with reward +5*
- *From  $s_2$ : action ‘stay’  $\rightarrow s_2$  with reward +2; action ‘go’  $\rightarrow s_1$  with reward -1*

*(a) Write the Bellman optimality equations for  $v_*(s_1)$  and  $v_*(s_2)$ .*

*(b) Solve for  $v_*(s_1)$  and  $v_*(s_2)$ .*

*(c) Determine  $\pi_*(s_1)$  and  $\pi_*(s_2)$ .*



**Problem 8** (Conceptual Questions). *Answer the following:*

- (a) *Why is the Bellman equation for  $v_*$  nonlinear while the Bellman equation for  $v_\pi$  is linear?*
- (b) *What does it mean for a state to have the Markov property?*
- (c) *Explain the difference between the reward signal and the value function. Why do we need both?*
- (d) *In GPI, the evaluation and improvement processes “compete.” Explain what this means.*
- (e) *Why might we prefer  $q_*$  over  $v_*$  for determining optimal actions?*

**Problem 9** (Computing  $q_\pi$  from  $v_\pi$ ). *Using the 3-state MDP from Problem 3 and the computed  $v_\pi$  values:*

(a) *Compute  $q_\pi(A, a_1)$ ,  $q_\pi(A, a_2)$ ,  $q_\pi(B, a_1)$ ,  $q_\pi(B, a_2)$*

(b) *Verify that  $v_\pi(A) = \sum_a \pi(a|A)q_\pi(A, a)$*

*Use the values:  $v_\pi(A) \approx 31.85$ ,  $v_\pi(B) \approx 25.74$ ,  $v_\pi(C) \approx 30.21$  (from Problem 3 solution).*

**Problem 10** (Action-Value Bellman Optimality). *For the 2-state MDP from Problem 7:*

- (a) *Write the Bellman optimality equations for  $q_*(s, a)$  for all state-action pairs.*
- (b) *Solve for all  $q_*$  values.*
- (c) *Verify that  $v_*(s) = \max_a q_*(s, a)$  for both states.*